Soil with Constant Surface Temperature modeled as Semi-Infinite Solid

Saturday 10th October, 2026

Curvenote

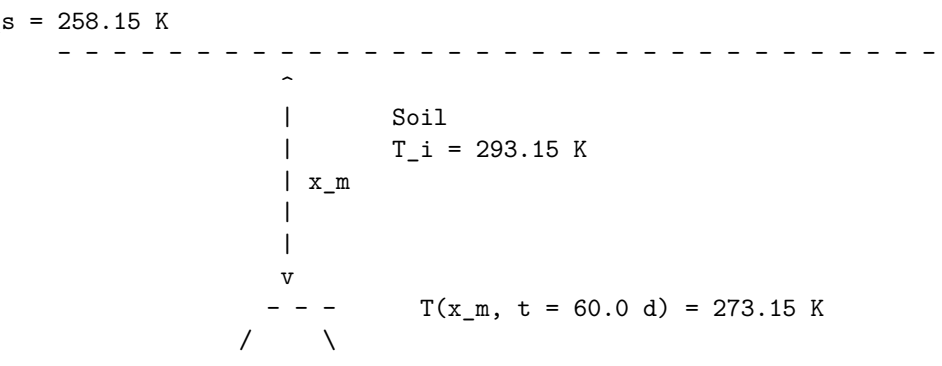
1 Problem Statement

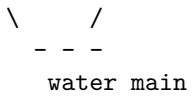
2 Assumptions

- One-dimensional conduction in x.
- Soil is a semi-infinite medium.
- Constant properties.

3 Schematic

- $\mathrm{k} = 0.52\ \frac{\mathrm{W}}{\mathrm{K\,m}}$
- $\mathrm{c} = 1840\ \frac{\mathrm{J}}{\mathrm{K\,kg}}$
- $\rho = 2050\ \frac{\mathrm{kg}}{\mathrm{m^3}}$
- $\alpha = 1.38 \times 10^{-7}\ \frac{\mathrm{m^2}}{\mathrm{s}}$





4 Analysis

4.1 Depth from surface, obtained from solving for x, given the closed-form equation for transient temperature distribution at constant surface temperature

x = $2.0\sqrt{t\alpha}\,\text{erfinv}\left(\frac{T-T_s}{T_i-T_s}\right)$ = 0.67696185 m

4.2 Depth to which soil has frozen

x_m = x = 0.67696185 m

5 Comments

- The properties of soil are highly variable, depending on the nature of the soil and its moisture content, and a representative range of thermal diffusivities is 1e-7 <α <3e-7 m2/s.

6 References

- F. P. Incropera, Fundamentals of heat and mass transfer. 6th ed. John Wiley & Sons, 2007, Ch. 5, sec. 5.7, p.288-290.